# The bandwidth of the floor-attaining census

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**Question.** The census ρ over frozen offsets (ξ_A, ξ_B) that attains Hall's floor is not separable in station coordinates, in sum/difference coordinates, or as a short mixture of either (versions 1–2; Results D–G). We ask instead how much structure it needs: for a census built only from harmonics e^{2i(k_Aξ_A + k_Bξ_B)} with |k_A|, |k_B| ≤ K, how closely can it reproduce the singlet correlators, and what does it pay above the floor when it can?

**Method.** For each (τ, s) cell the census over the 1024 offsets is a trigonometric polynomial of degree K with free coefficients. Non-negativity is imposed at every atom; normalization, the correlator band (tolerance ε), the CHSH row and the measurement-dependence rows are exactly those of `dither_sweep_rho.py`. Two convex LPs per K (HiGHS interior point, dual simplex as fallback): the *price* LP minimizes MD subject to the band; the *miss* LP minimizes the band width any K-band census needs. Both solutions are re-verified from the reconstructed census (positivity, mass, floor, band). "Infeasible at ε" means the certified minimum miss exceeds ε. Positivity at the atoms, not on the continuum, is the discretization; the unrestricted census is itself representable at K ≈ 16, so the price must reach 1.000 there by construction. Scripts: `census_bandlimit_v1–v4.py`, launchers `run_parallel_v1–v3.py`; the v4 certificates supersede the "infeasible" labels of v1–v3, which were read from a solver status code.

**Minimum miss of the best K-band census** (the smallest correlator tolerance a K-band census can reach), random0; K = 5 on all four censuses:

| K | 2 | 3 | 4 | 5 | 5 (random1) | 5 (strat0) | 5 (strat1) |
|---|---|---|---|---|---|---|---|
| min miss | — (see i) | 0.1052 | 0.0174 | 0.0120 | 0.0120 | 0.0131 | 0.0110 |

**Price (MD/floor) of the smoothest attaining census**, ε = 0.01, random0; K = 6 replicated on four censuses (random1 2.094, strat0 2.019, strat1 2.081):

| K | ≤ 5 | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|---|
| MD/floor | — (miss > 0.01) | 2.274 | 1.861 | 1.598 | 1.356 | 1.121 |

**The (ε, K) surface**, random0 (unrestricted price 1.000 at every ε):

| K ε | 0.01 | 0.02 | 0.05 | 0.10 |
|---|---|---|---|---|
| 6 | 2.274 | 1.913 | 1.590 | 1.451 |
| 8 | 1.861 | 1.645 | 1.389 | 1.275 |
| 10 | 1.598 | 1.428 | 1.230 | 1.132 |

The skeleton (leading pixels (2,2), (2,0), (2,4), (1,1), (0,2), (2,−2), (1,−1), (3,−1) and their phases) is the same at every feasible K.

**Statements.** (i) On this terrain, reproducing the singlet to 1% requires K ≥ 6: the best K = 5 census misses by 1.1–1.3% on four independent censuses, K = 4 by 1.7%, K = 3 by 10.5% (certified LP minima, not search outcomes); at K = 2 no non-negative census satisfies the CHSH row at all. (ii) The smoothest attaining census at ε = 0.01 pays 2.0–2.3× Hall's floor across four censuses; the excess falls roughly linearly in K (≈ 0.12 per harmonic) and is sampling-limited beyond K ≈ 14. (iii) The smoothness price softens with tolerance: at K = 6, 127% over the floor at ε = 0.01, 91% at 0.02, 59% at 0.05, 45% at 0.10; the surface is monotone in both variables. (iv) The floor requires a census with angular features of order 6°; a smooth birth pays above the floor by the surface's value.

**Also in this version.** *Result G* — rank-k mixtures of (S, D)-separable censuses, ranks 1–3, 8 starts each: none feasible at ε = 0.01; misses 2.65 → 2.46 → 2.37%. *Result H* — pooled (S, D) spectrum of the four censuses: eleven cross pixels above the permutation null; S preferred at 39.5° mod 90°; D tracks S inside the lobes. *Result M (solve frame)* — re-solving random0's offsets with Bob's setting pinned and Alice's swept (`dither_sweep_swap_v1.py`) attains the same floor (md = 0.138071187458) and the same census: (2,2) at φ = 157.4° against ≈157° in the pinned frame, preferred sum 39.35° against 39.5°, the same (S, D) cross pixels and both families of push. The station asymmetry of the census ((2,0) = 0.26 vs (0,2) = 0.13) stays on the same side, so it is a property of the terrain, not of the solve frame; whether the terrain's own asymmetry is physical or inherited from the breeding run (which evaluated fitness with Alice pinned) remains open.

**Scope.** One terrain (the paper's greedy terrain), four censuses plus one roles-swapped solve, 1024-offset banks; the K ≥ 14 tail is set by the sampling; feasibility and infeasibility are both certified by LP optima re-verified from the census. Every number above is copied from the deposited transcripts (`parallel_console.txt`, `parallel2_console.txt`, `parallel3_console.txt`, `census_bandlimit_v2_random0_10_22_26.txt`, `census_swap_random0_10_23_26.txt`, `sumdiff_swap_random0_10_23_26.txt`).